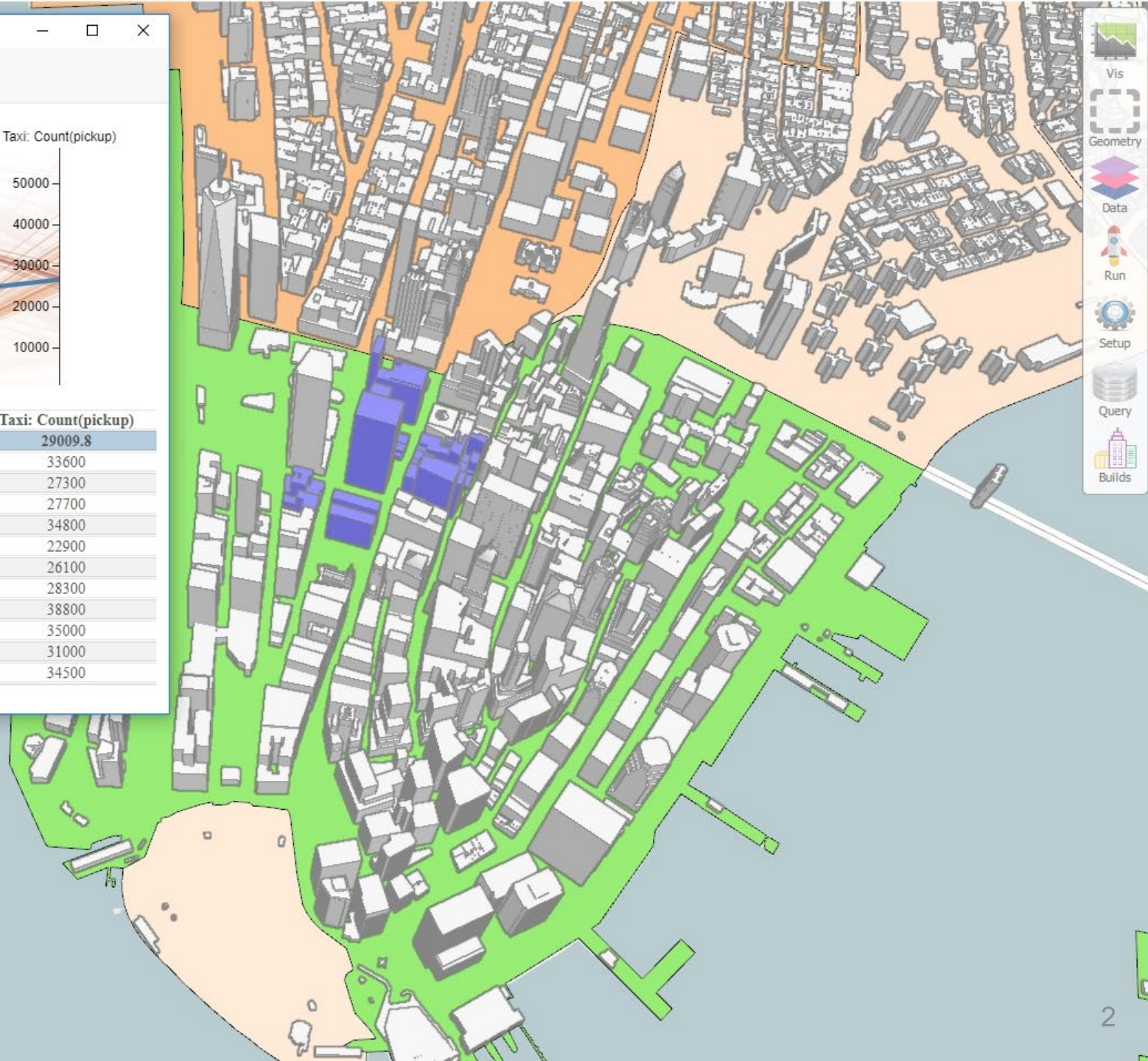
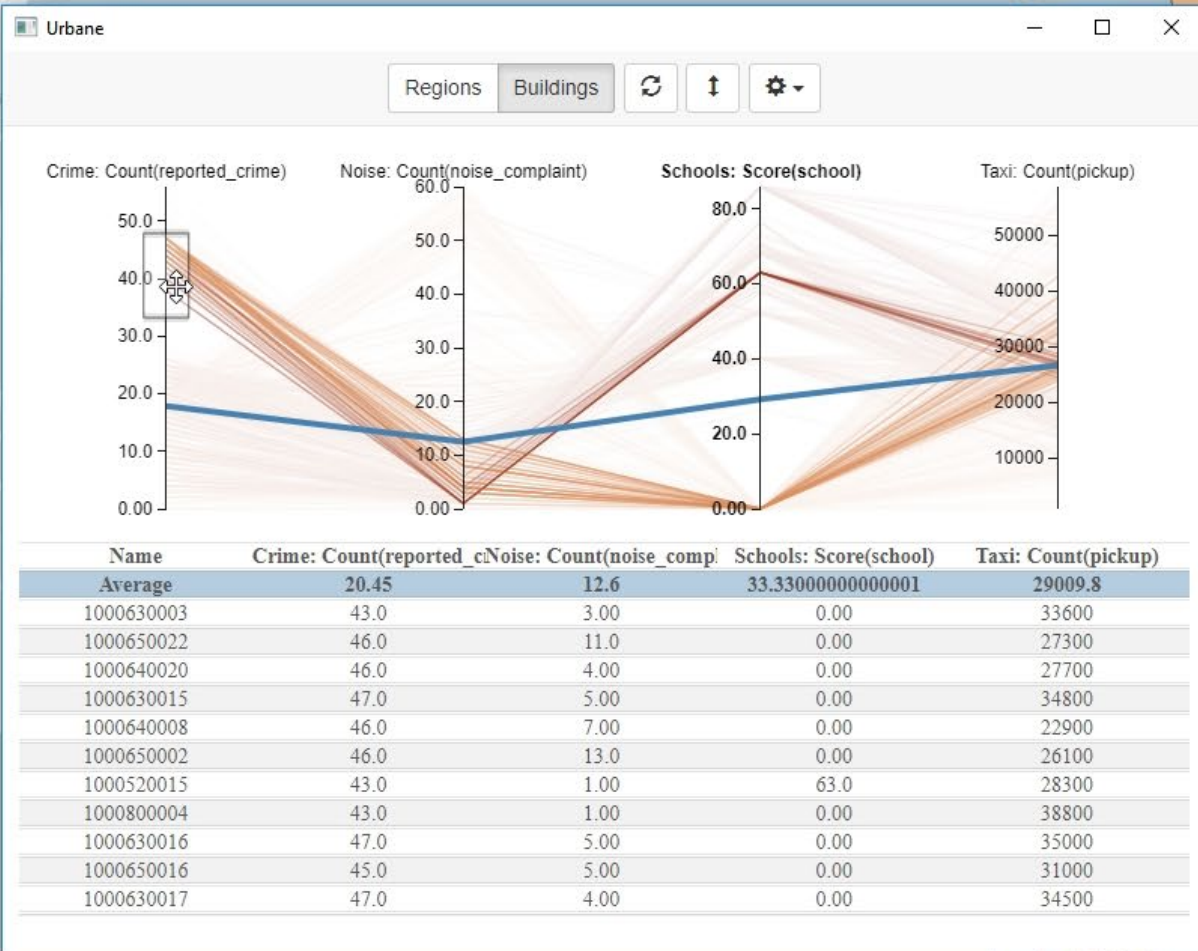


Visual Analytics

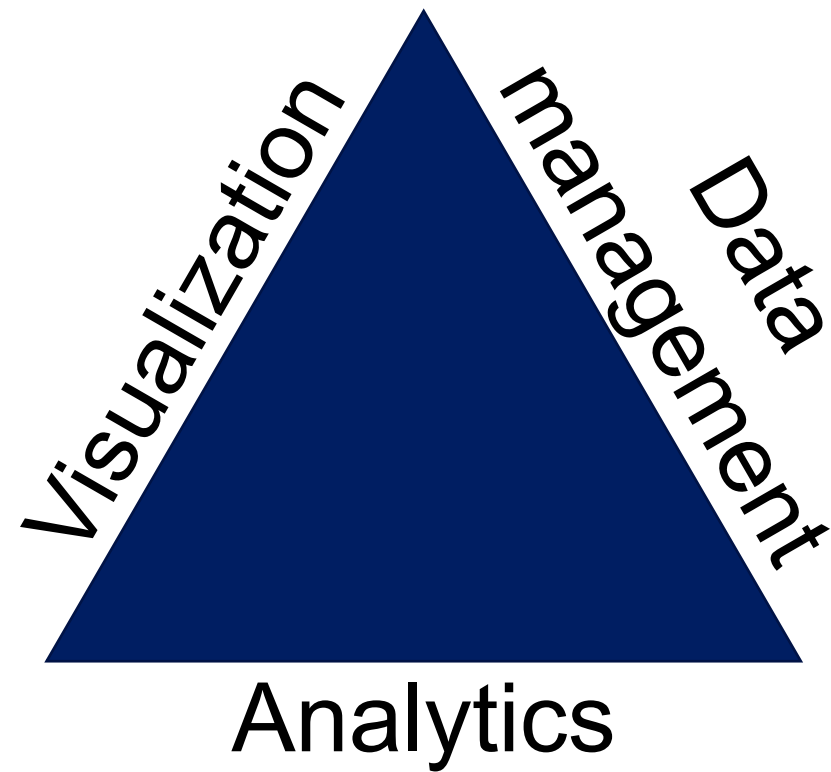
CS424: Visualization & Visual Analytics

Fabio Miranda

<https://fmiranda.me>



Visual Analytics



Visual Analytics



“Visual analytics is the science and practice of analytical reasoning by combining computational processing with visualisation. These are tightly-coupled using interactive techniques so that each informs the other.”

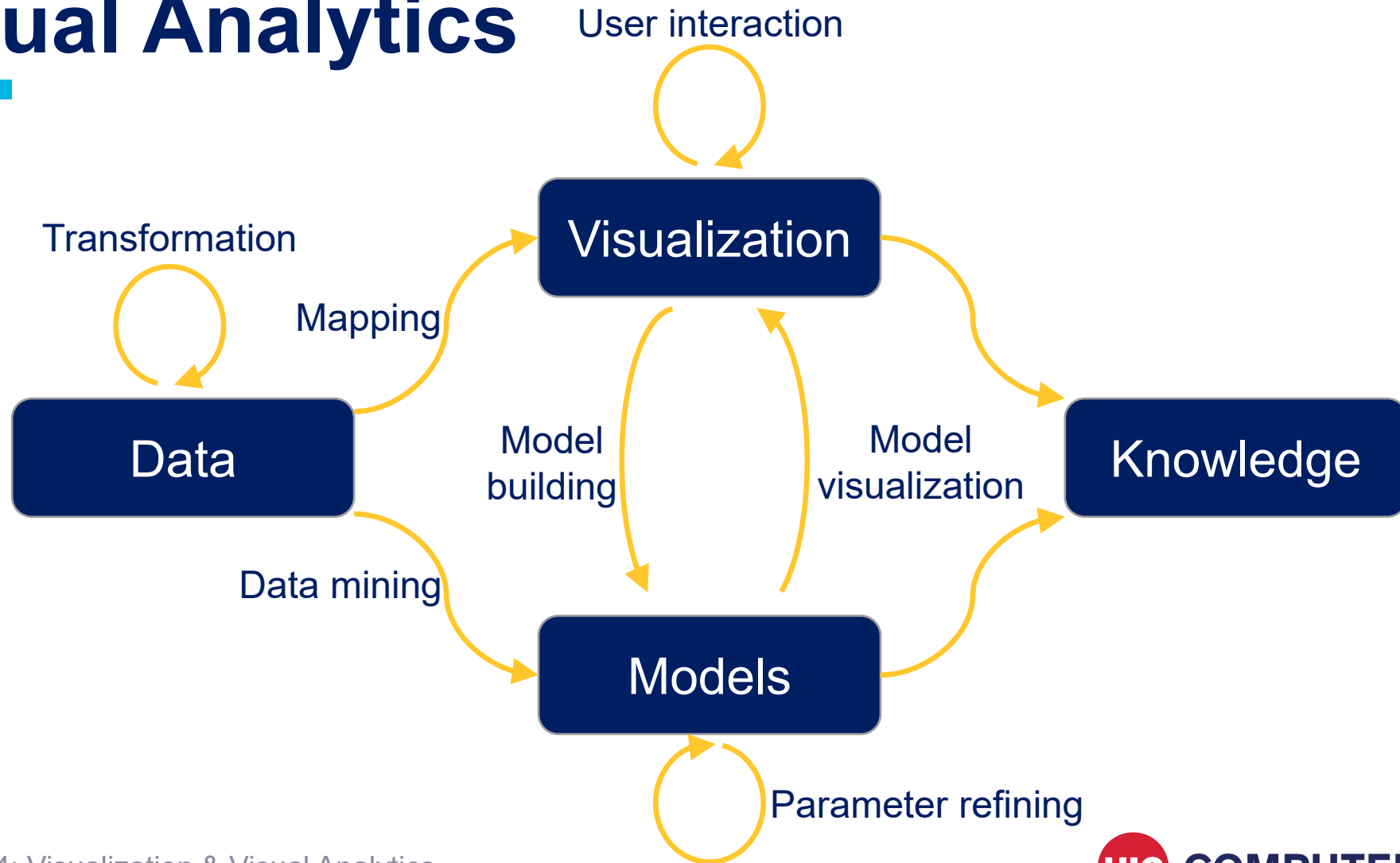
Visual Analytics

“Visual analytics is the science and practice of analytical reasoning by combining computational processing with visualisation. These are tightly-coupled using interactive techniques so that each informs the other.”

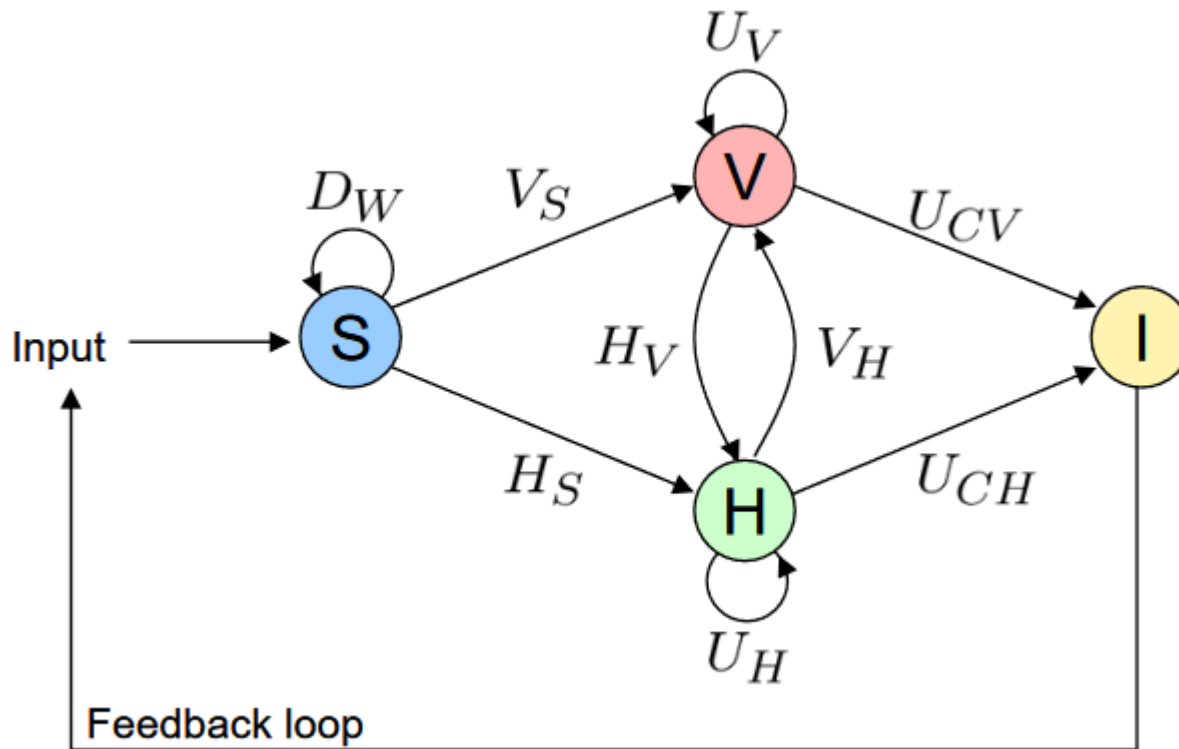
“The involvement of computational processing allows **computers to do what computers are good at**: transforming and summarising data and searching for specific pieces of information. Interactive visual interfaces involve the human analyst, **allow him or her to do what human analysts are good at**: interpreting and reasoning.”

[Andrienko et al., Introduction to Visual Analytics by an Example]

Visual Analytics



Visual analytics process



[“Visual Analytics: Scope and Challenges”, Kleim et al., 2008]

$$F: S \rightarrow I$$

$$f \in \{D_w, V_x, H_y, U_z\}$$

S : data, I : insight

D_w : data pre-processing functionalities (data transformation, data cleaning, data selection, data integration)

$V_x, x \in \{S, H\}$: visualization functions

V_S : functions visualizing data ($S \rightarrow V$)

V_H : functions visualizing hypotheses ($H \rightarrow V$)

$H_y, y \in \{S, V\}$: hypotheses generation process

H_S : hypotheses from automatic process ($S \rightarrow H$)

H_v : hypotheses from visualization ($V \rightarrow H$)

$U_z, z \in \{V, H, CV, CH\}$: user interactions

U_V : visualization interaction ($V \rightarrow V$)

U_H : hypotheses interaction ($H \rightarrow H$)

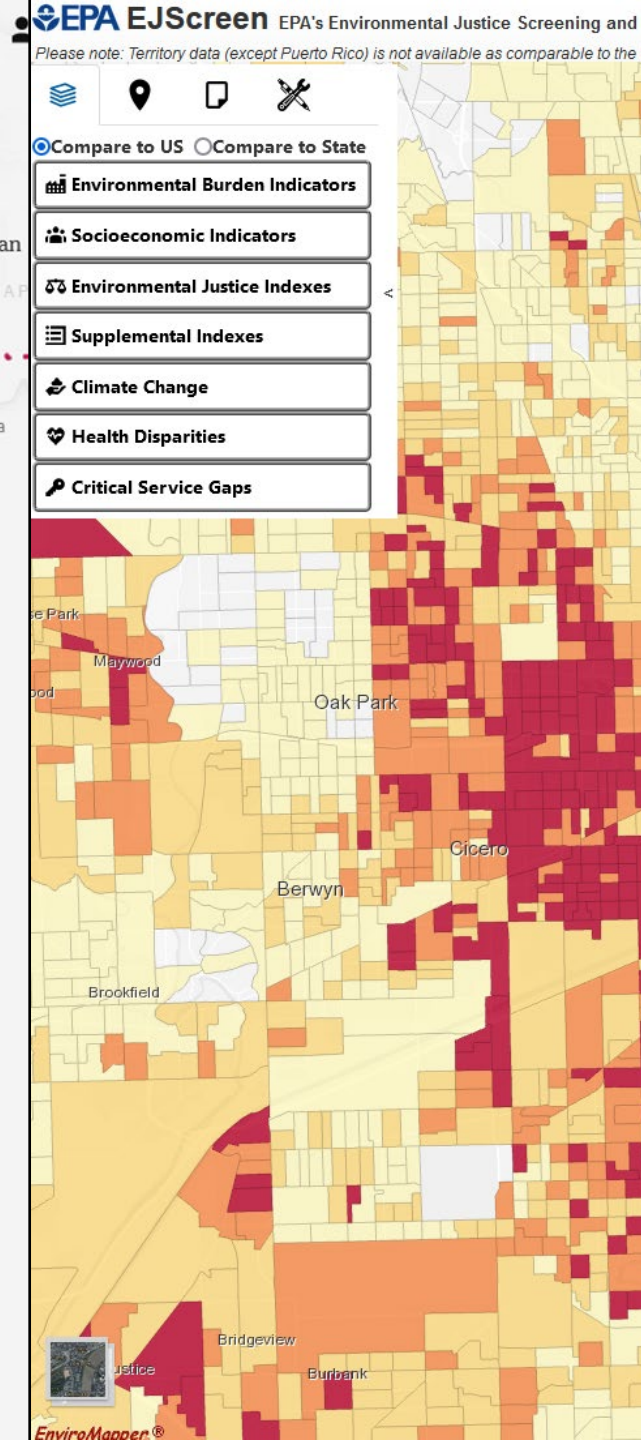
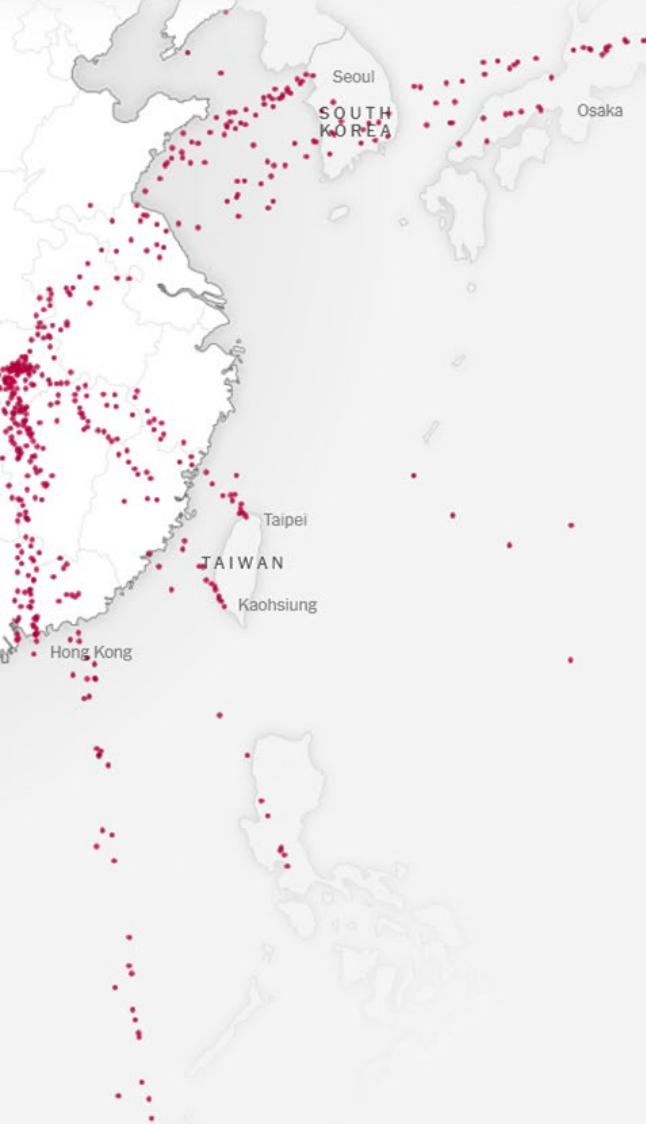
U_{cv} : insight concluded from visualization

U_{ch} : insight concluded from hypothesis

How the Virus Got Out

By Jin Wu, Weiyei Cai, Derek Watkins and James Glanz March 22, 2020

The most extensive travel restrictions to stop an outbreak in human history haven't been enough. We analyzed the movements of hundreds of millions of people to show why.



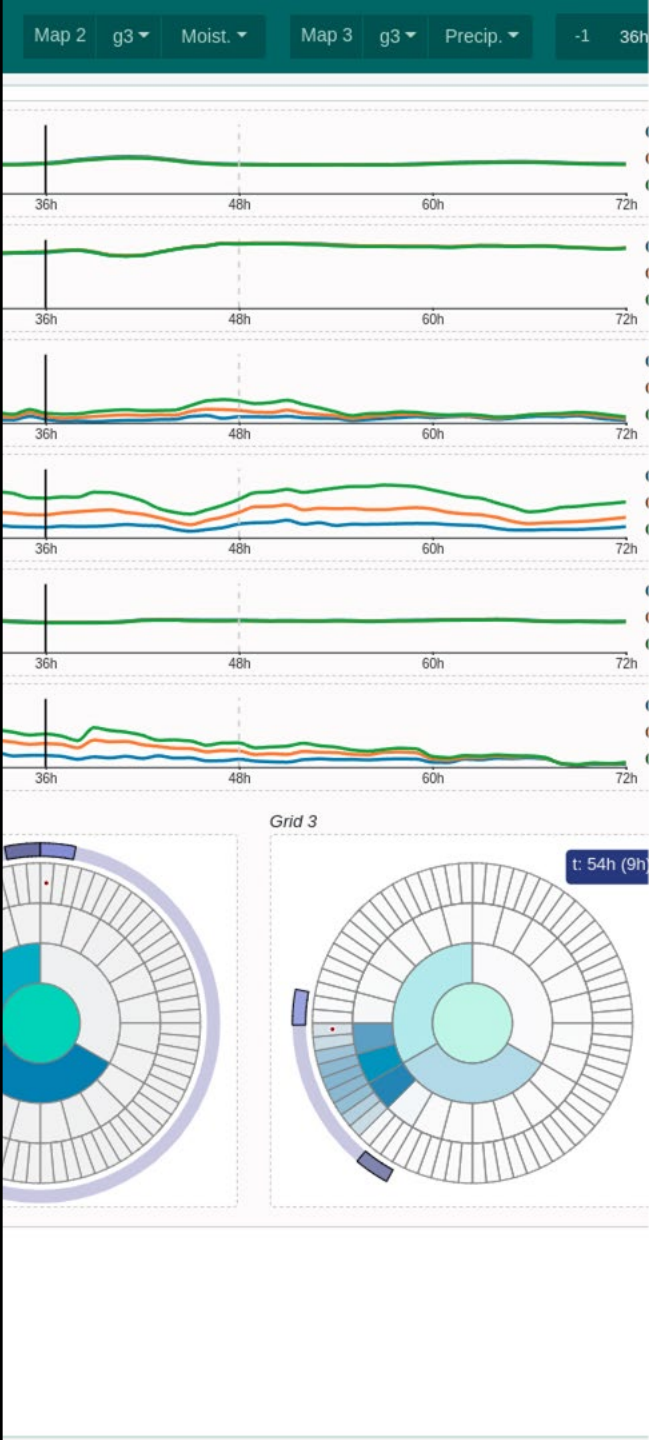
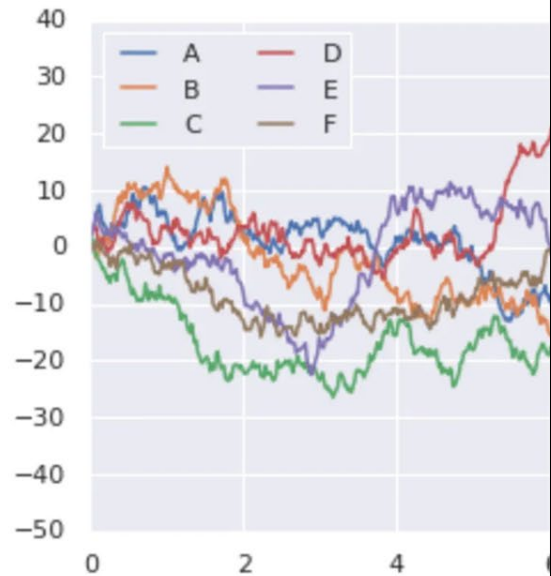
```
[5]: import matplotlib.pyplot as plt
plt.style.use('classic')
%matplotlib inline
import numpy as np
import pandas as pd
import seaborn as sns
sns.set()
```

```
[6]: rng = np.random.RandomState(0)
x = np.linspace(0, 10, 500)
y = np.cumsum(rng.randn(500, 6))
```

Next step

Now, create a graph.

```
[7]: plt.plot(x, y)
plt.legend('ABCDEF', ncol=2, loc='upper left')
```

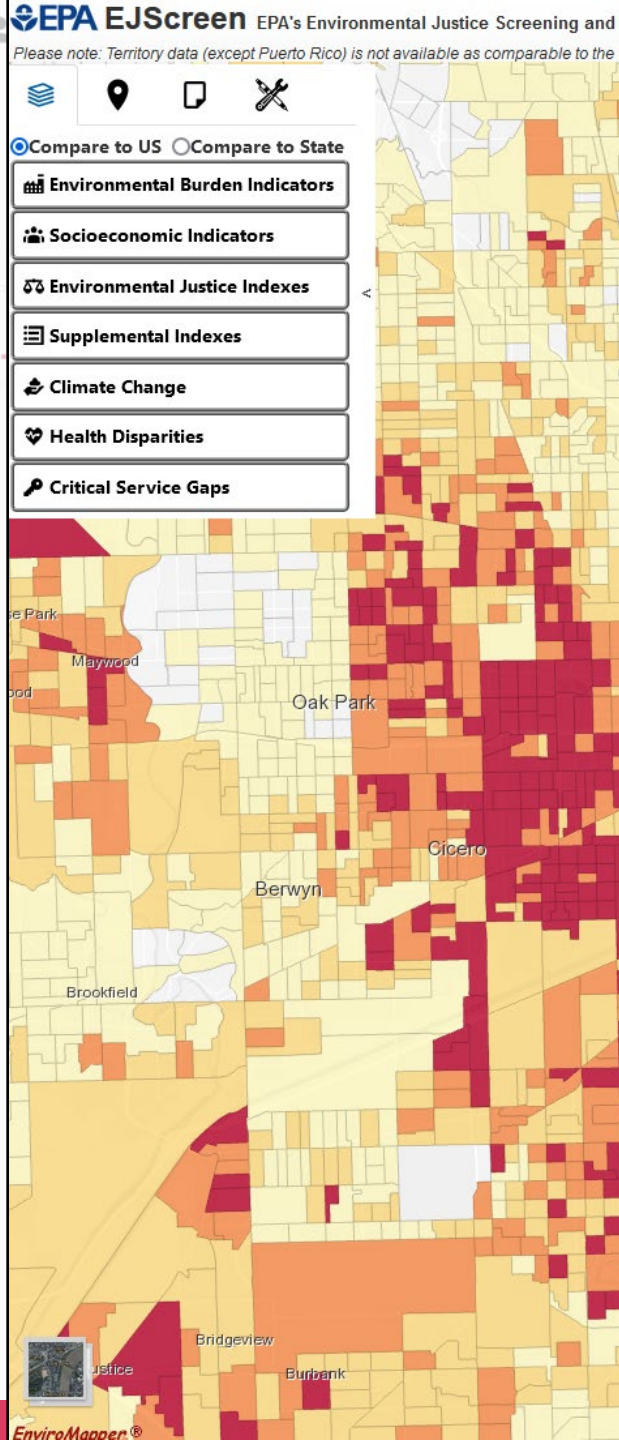


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Data journalism



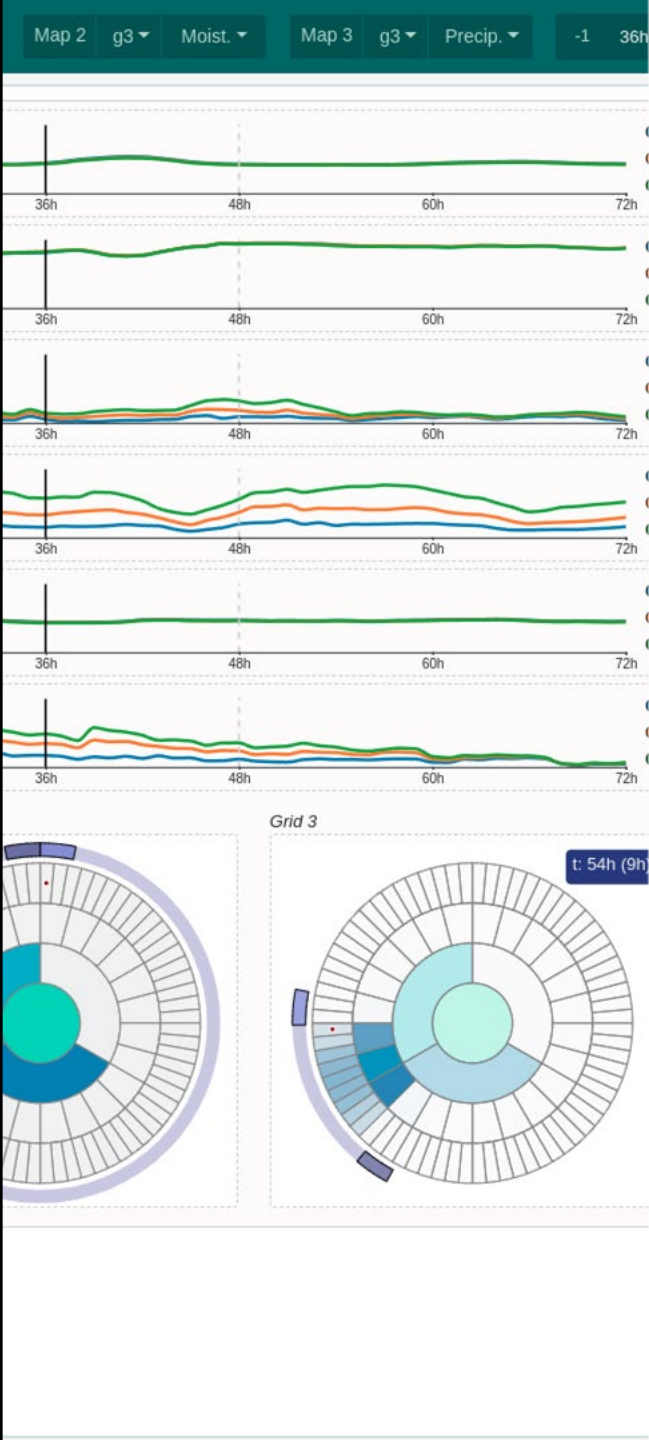
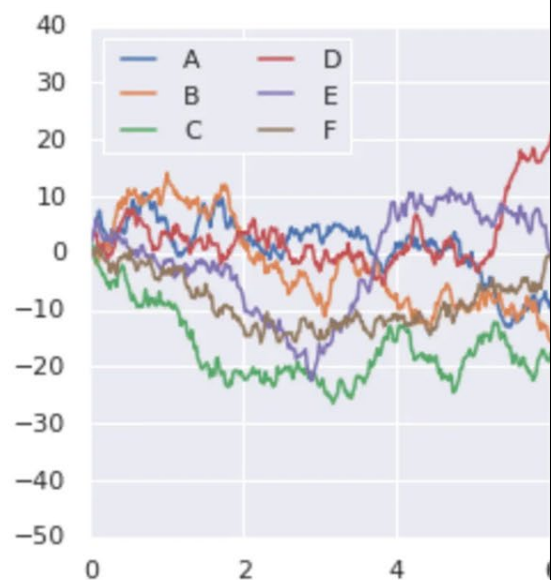
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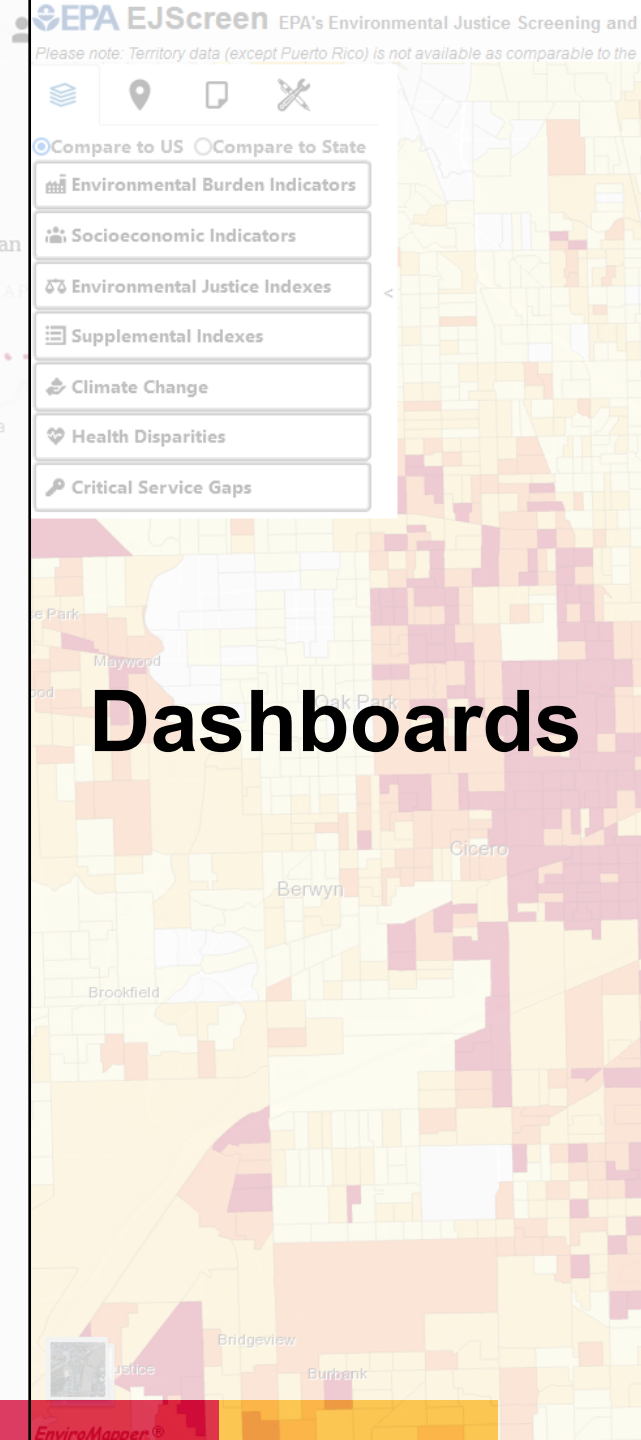
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Data journalism



Dashboards

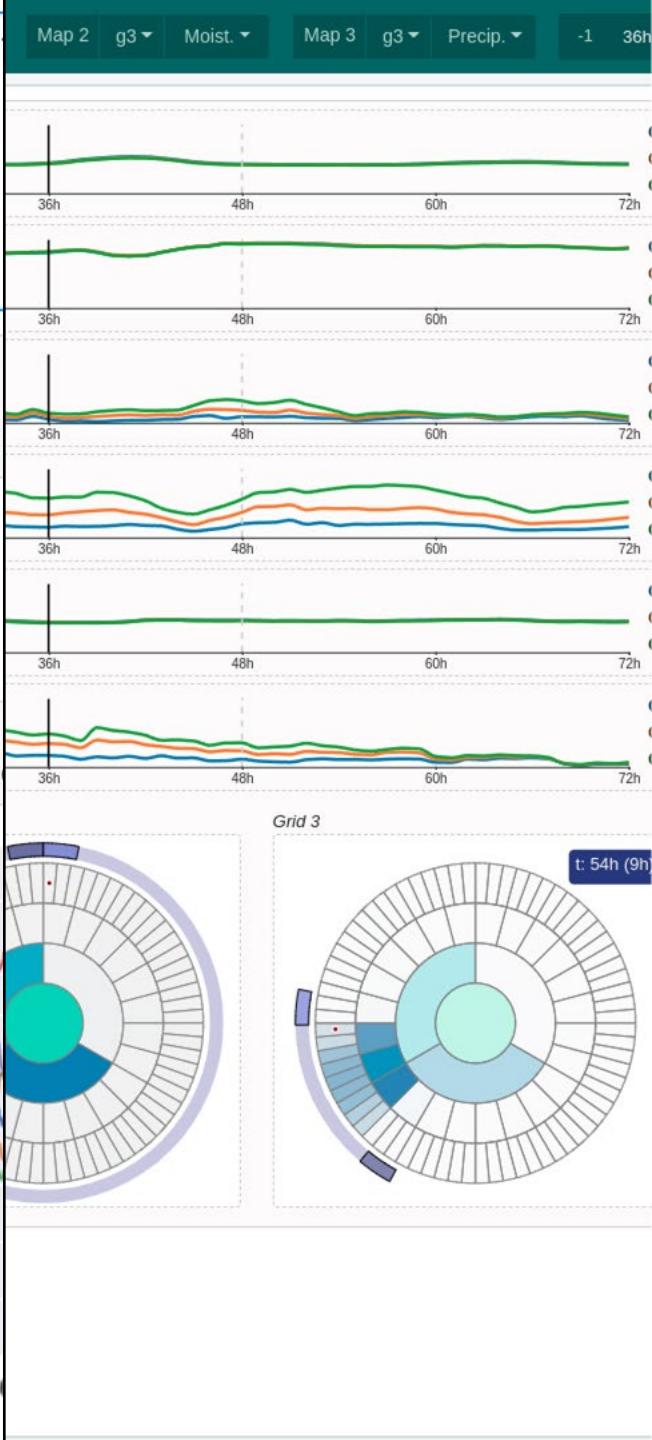
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The New York Times

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Data journalism

EPA EJScreen

EPA's Environmental Justice Screening and Reinvestment Tool

Please note: Territory data (except Puerto Rico) is not available as comparable to the United States.

Compare to US

Compare to State

Environmental Burden Indicators

Socioeconomic Indicators

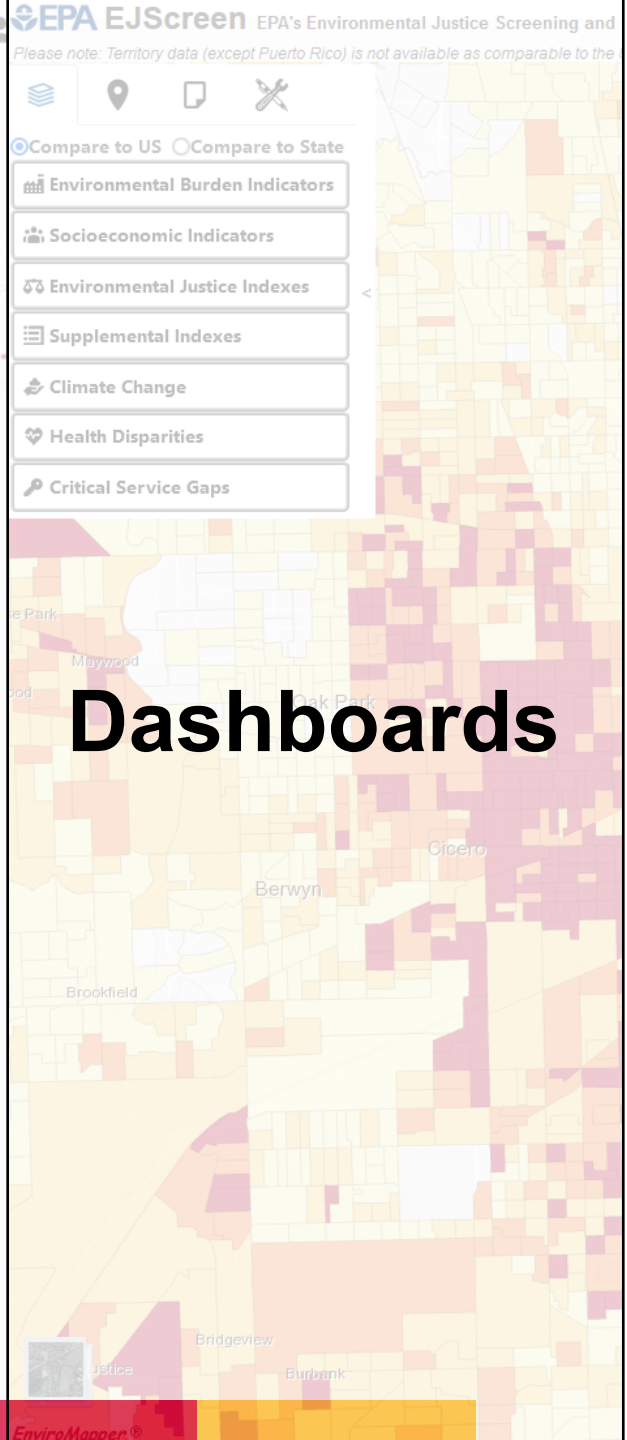
Environmental Justice Indexes

Supplemental Indexes

Climate Change

Health Disparities

Critical Service Gaps



Dashboards

[5]:

```
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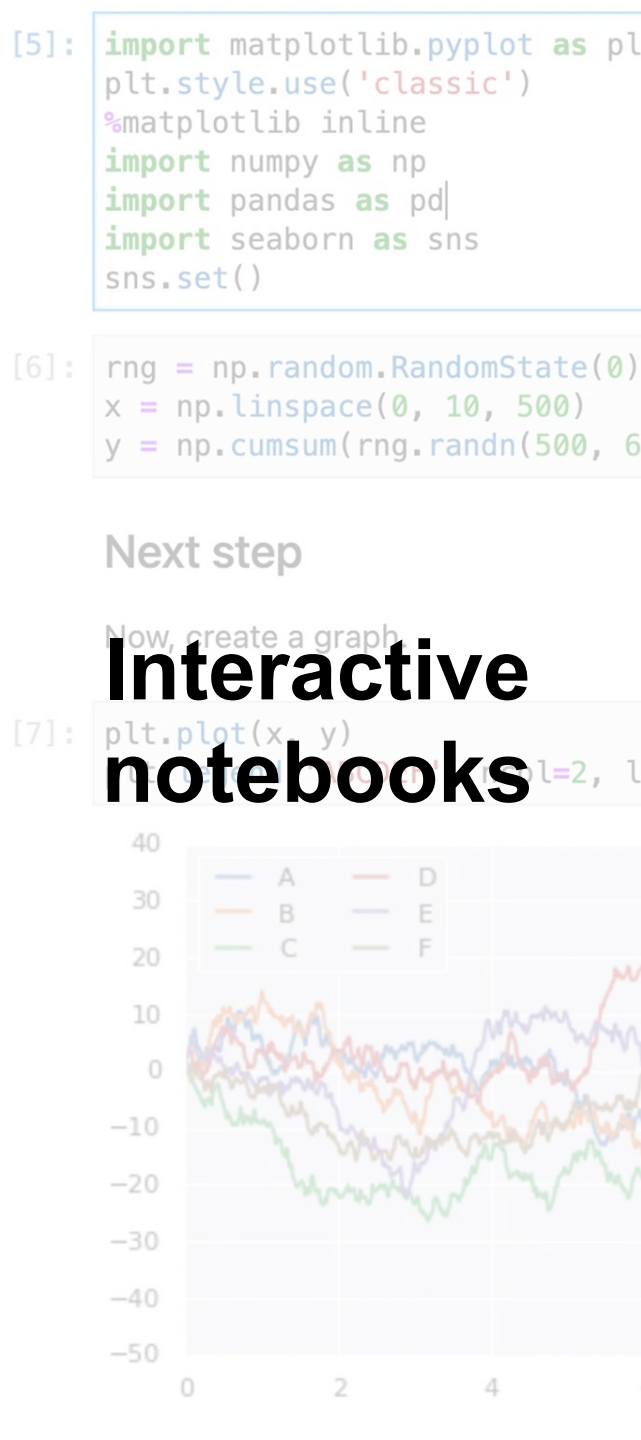
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rng = np.random.RandomState(0)
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Next step

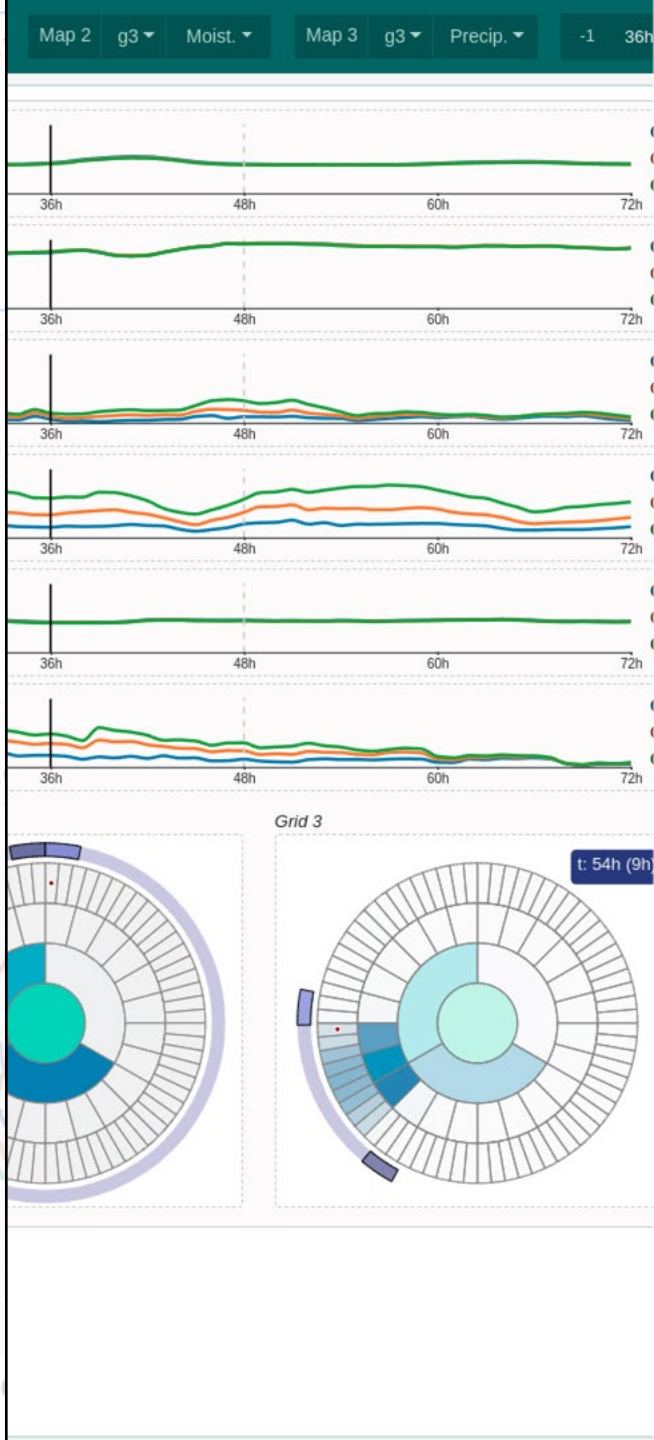
Now, create a graph

[7]:

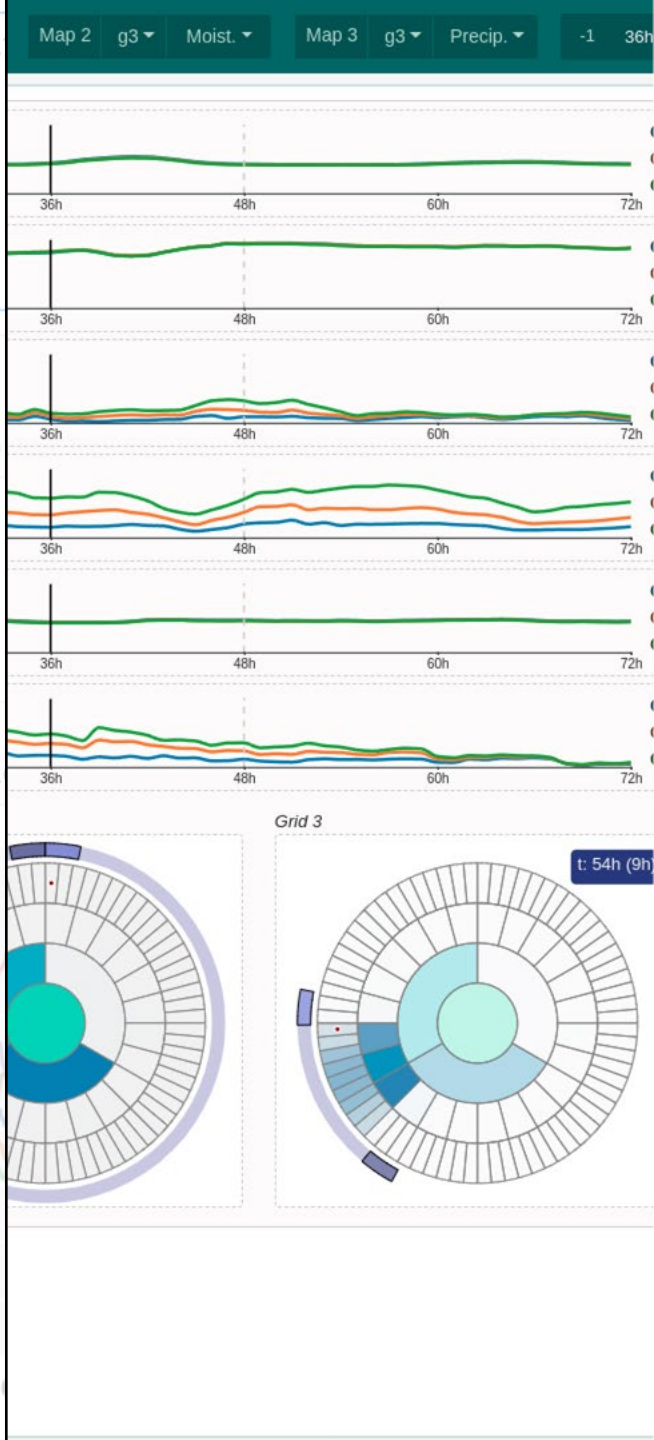
```
plt.plot(x, y)
```



Map 2 g3 Moist. Map 3 g3 Precip. -1 36h



Grid 3



The New York Times

How the Virus Got Out

By Jin Wu, Weiyi Cai, Derek Watkins and James Glanz March 22, 2020

The most extensive travel restrictions to stop an outbreak in human history haven't been enough. We analyzed the movements of hundreds of millions of people to show why.



A map showing travel movements between East Asia (including Seoul, Osaka, Taipei, Kaohsiung, and Hong Kong) and North America. Red dots represent travel paths and destinations.

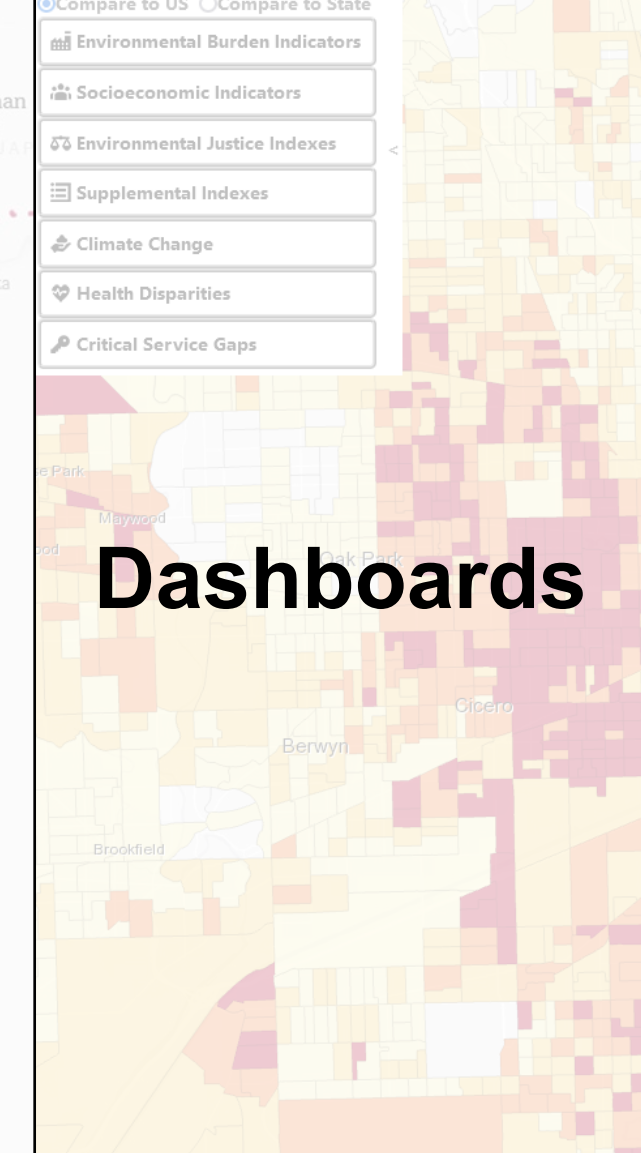
Data journalism

EPA EJScreen EPA's Environmental Justice Screening and Reinvestment Tool

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Compare to US Compare to State

- Environmental Burden Indicators
- Socioeconomic Indicators
- Environmental Justice Indexes
- Supplemental Indexes
- Climate Change
- Health Disparities
- Critical Service Gaps



A heatmap showing environmental justice indicators across Chicago neighborhoods. The map is color-coded from yellow (low burden) to red (high burden). Neighborhoods like Cicero, Berwyn, and Bridgeview are labeled.

Dashboards

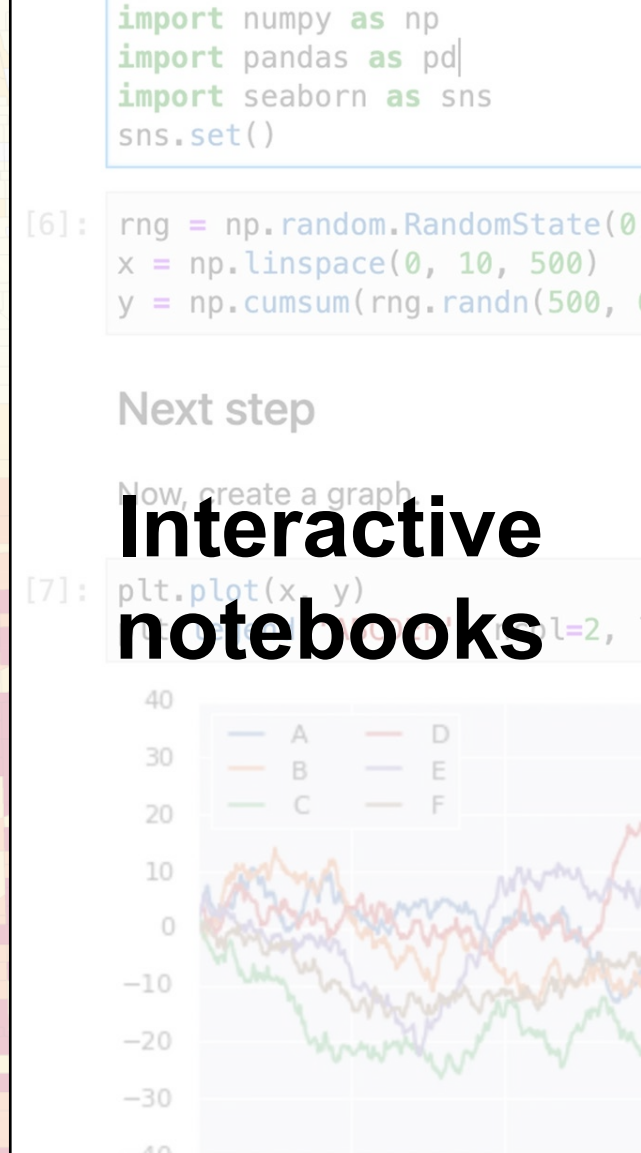
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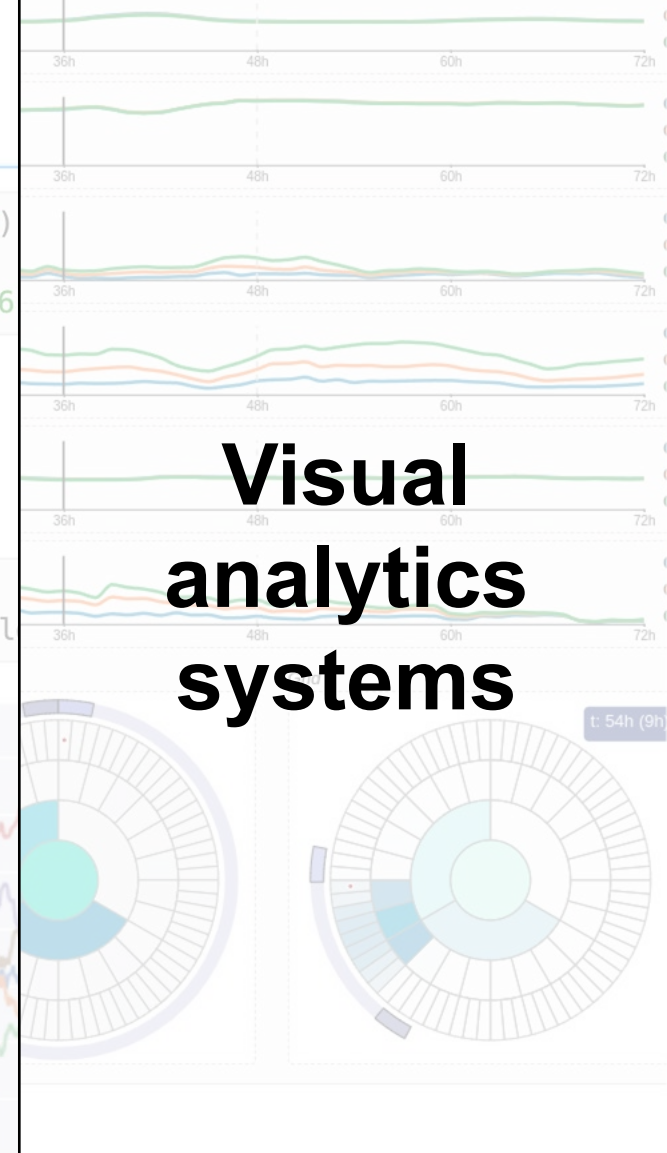
```
[7]: plt.plot(x, y)
```



A line plot showing the cumulative sum of random noise over time. The x-axis ranges from 0 to 5, and the y-axis ranges from -50 to 40. Multiple lines are plotted, representing different random states.

Interactive notebooks

Map 2 g3 Moist Map 3 g3 Precip -1 36h



A series of time-series plots showing data over 72 hours. The plots are arranged in a grid, with each plot showing a different variable. Below the plots are two sunburst charts, which are used for hierarchical data visualization.

Visual analytics systems

Visual analytics mantra

- Visualization mantra:

“Overview first, zoom / filter, details on demand”

[Shneiderman, 1996]

- Visual analytics mantra:

“Analyze first - show the important - zoom, filter and analyze further - details on demand”

[Kleim et al., 2008]

Dashboards and visual analytics systems (Conrow et al., 2023)

- “Both visual analytics (VA) and dashboards facilitate analytical reasoning and decision-making by simplifying problems and communicating results visually. However, a dashboard display differs from a VA platform along three key aspects”:
 - “VA platforms are information processing systems that can handle raw, unprocessed, and unstructured data from heterogeneous sources, whereas dashboards provide a user interface that is based on data that are often already standardized (i.e. cleaned and integrated) and transformed.”
 - “The second key difference between VA and dashboards relates to their analytical complexity. The tools and analytical processes that produce insights from large, complex datasets lie specifically within the VA domain and outside the scope of typical dashboard applications, which focus on informational displays rather than analyzing and manipulating input data.”
 - “This leads to the third key difference between VA and dashboards: user interaction and output display. Dashboards emphasize at-a-glance views and monitoring for only relevant data points, whereas VA is meant to provide deep understanding and knowledge discovery.

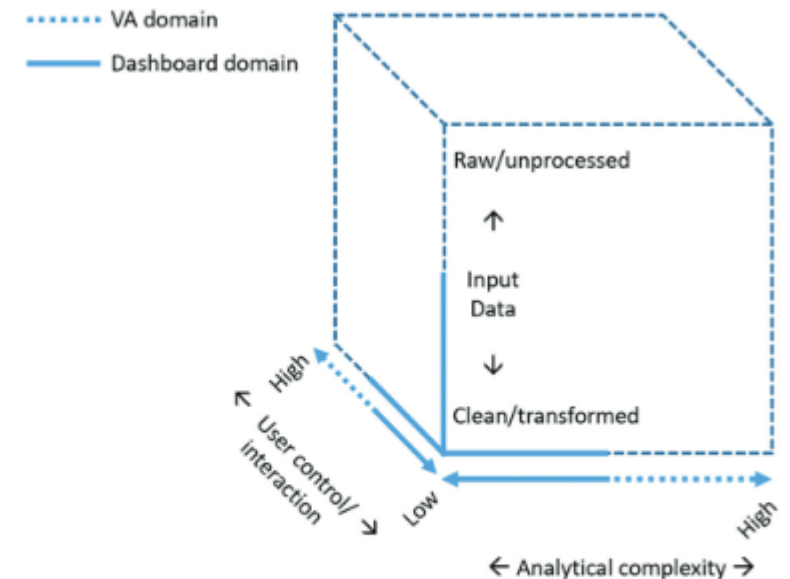
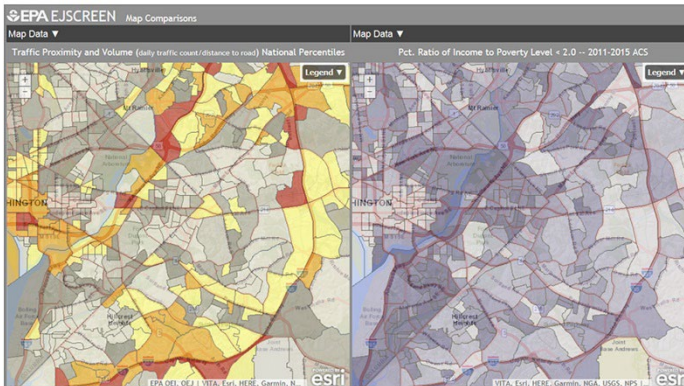


Figure 1. Comparison of visual analytics platforms and dashboards in the three key dimensions: input data, analytical complexity, and user interaction for output display.



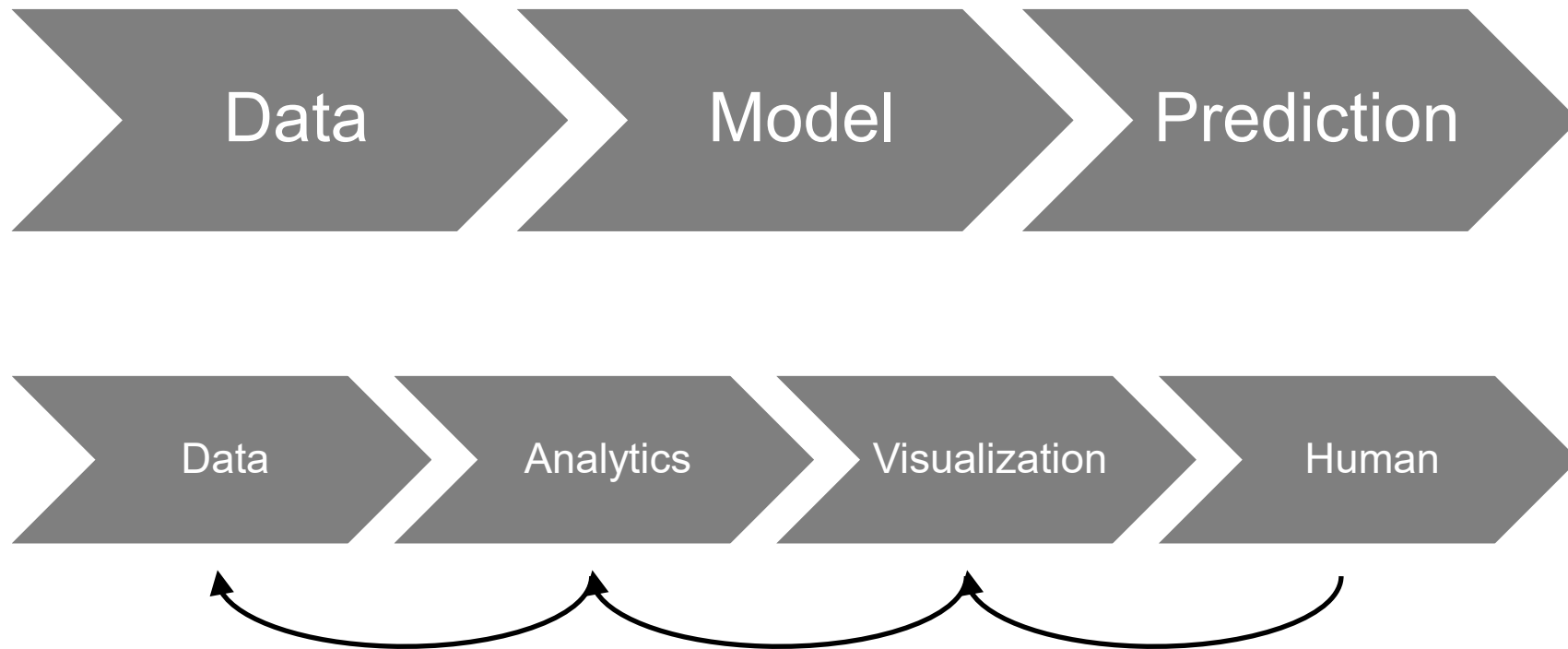
VA systems

Think of VA systems as *pipelines of human–machine reasoning*

VA systems



VA systems



VA systems

Role	Description	“Just AI”
Exploratory	Enables open-ended discovery, pattern seeking, hypothesis generation	AI requires pre-defined objectives or loss functions
Explanatory	Externalizes intermediate reasoning (e.g., feature importance, uncertainty)	AI outputs opaque numeric models
Integrative	Merges heterogeneous data, spatial context, qualitative factors	AI tends to silo modalities or discard context

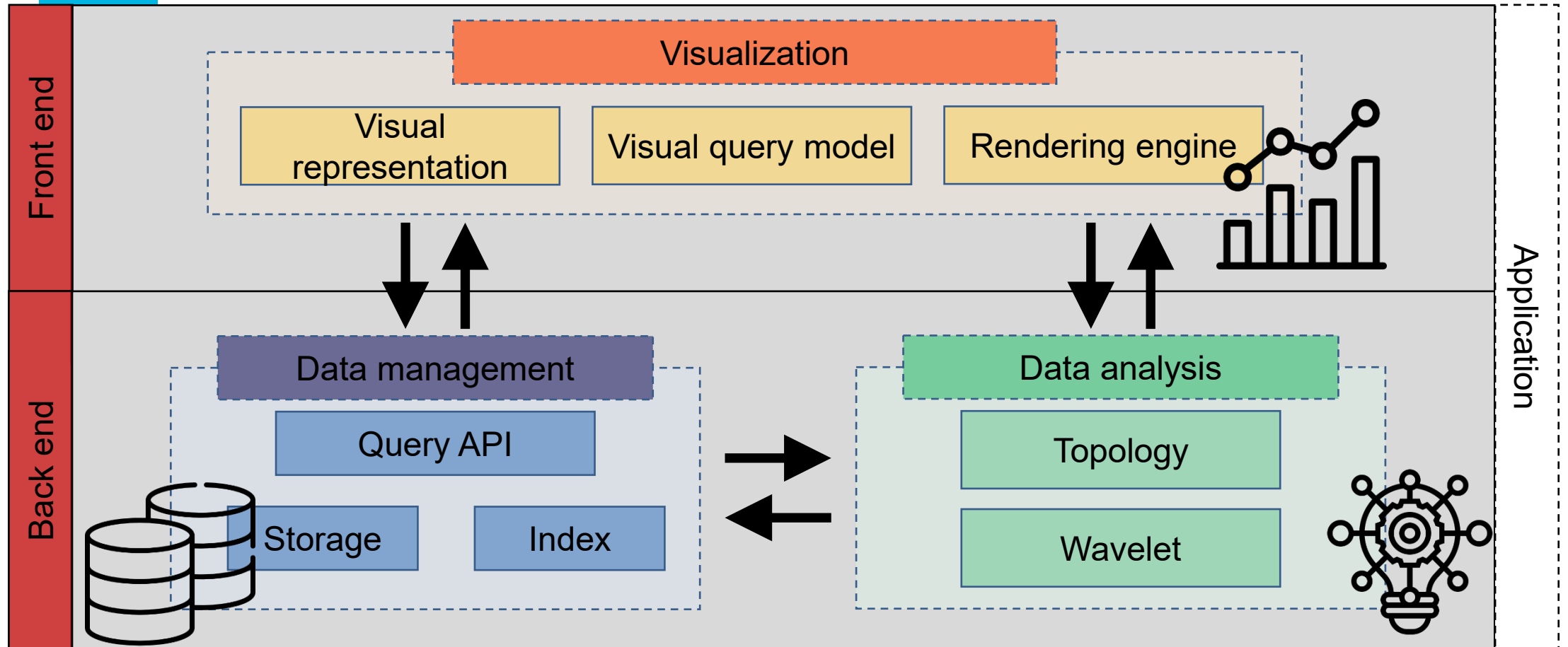
VA systems



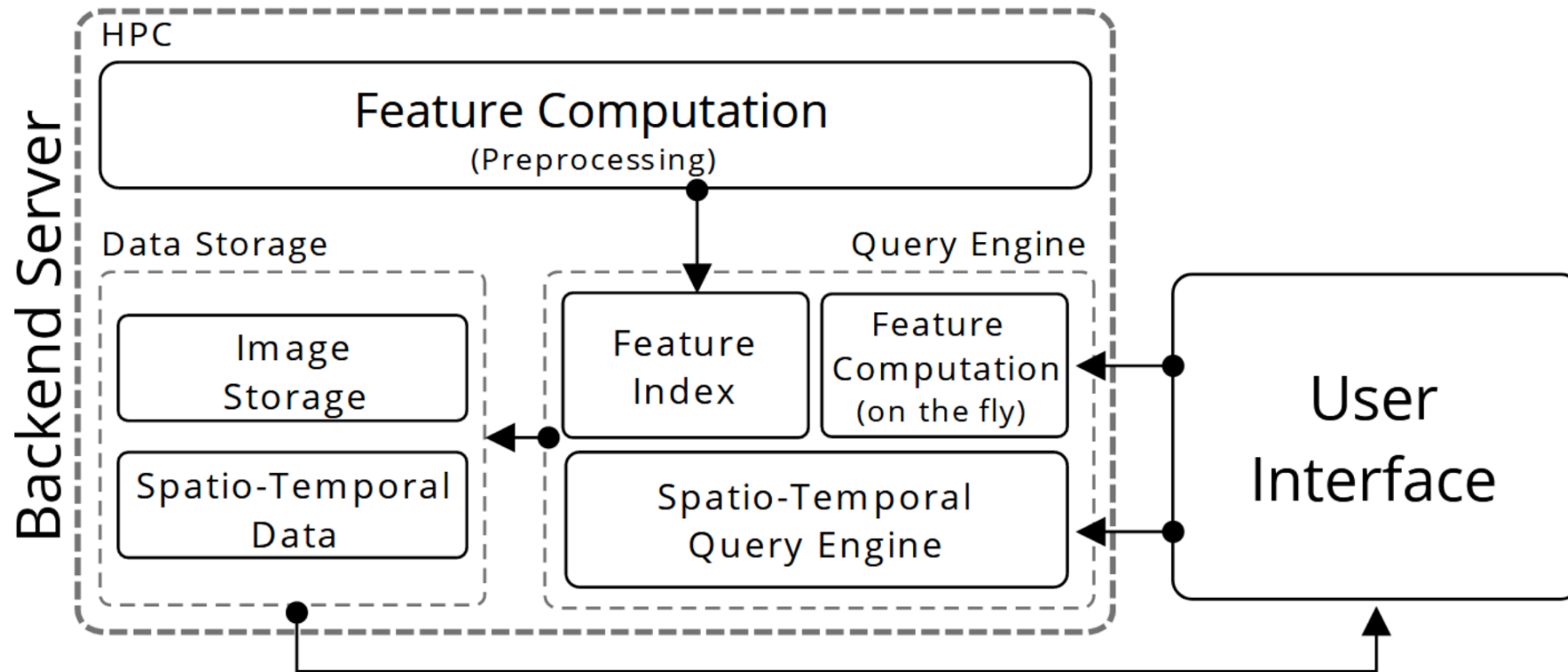
AI automates inference; visualization amplifies human sensemaking

Visualization in visual analytics is not about showing what the AI found – it's about steering, contextualizing and validating these findings.

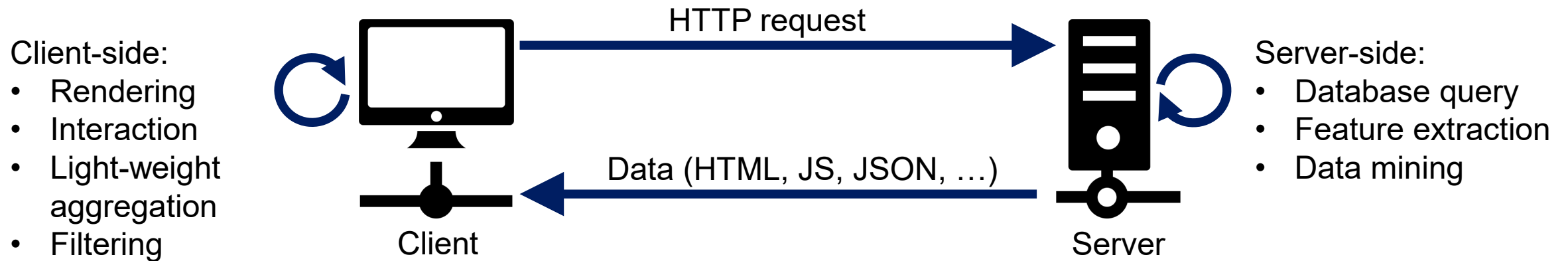
A visualization system



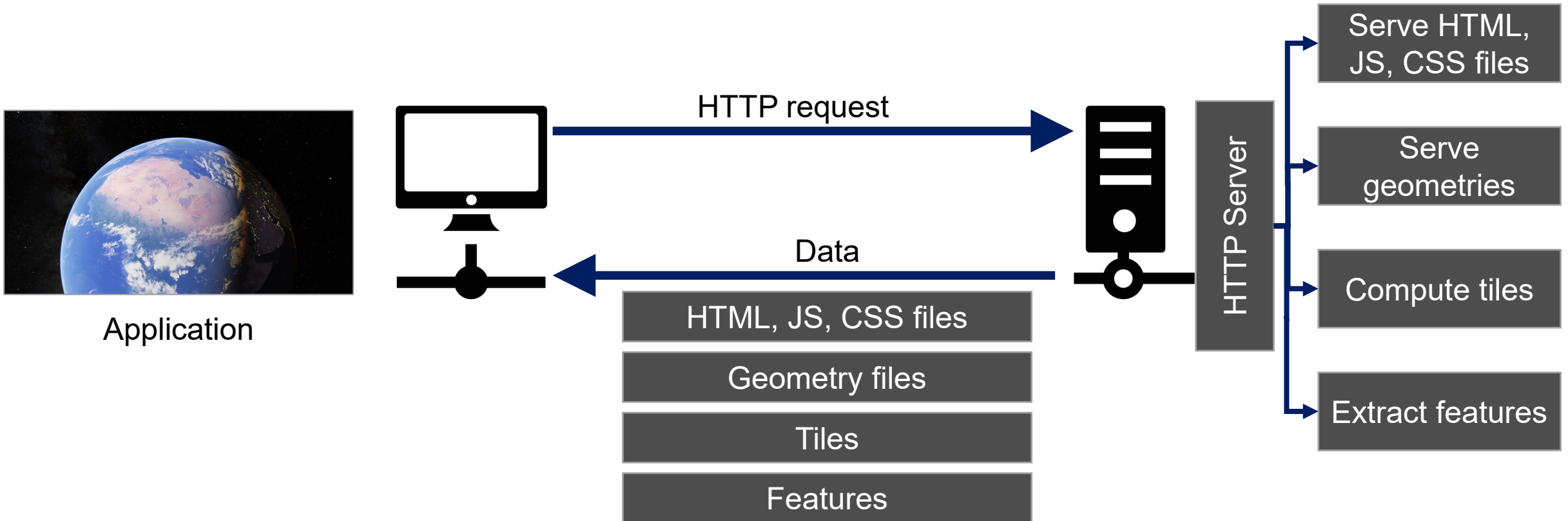
A visualization system example



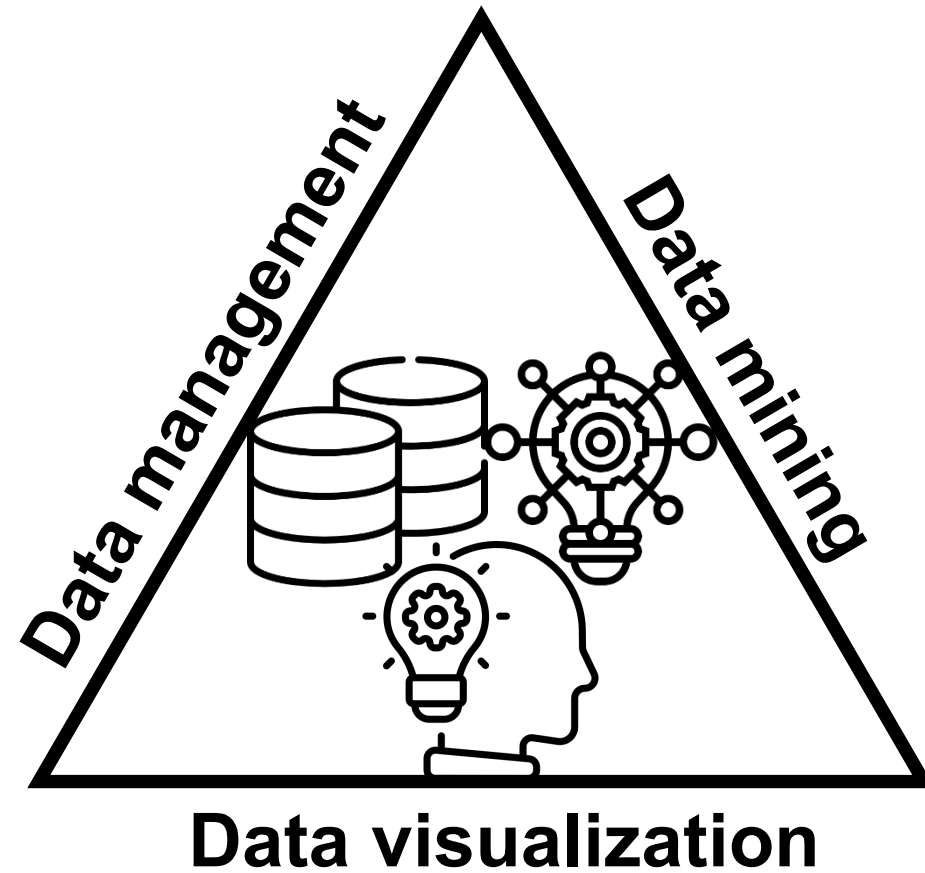
Client and server



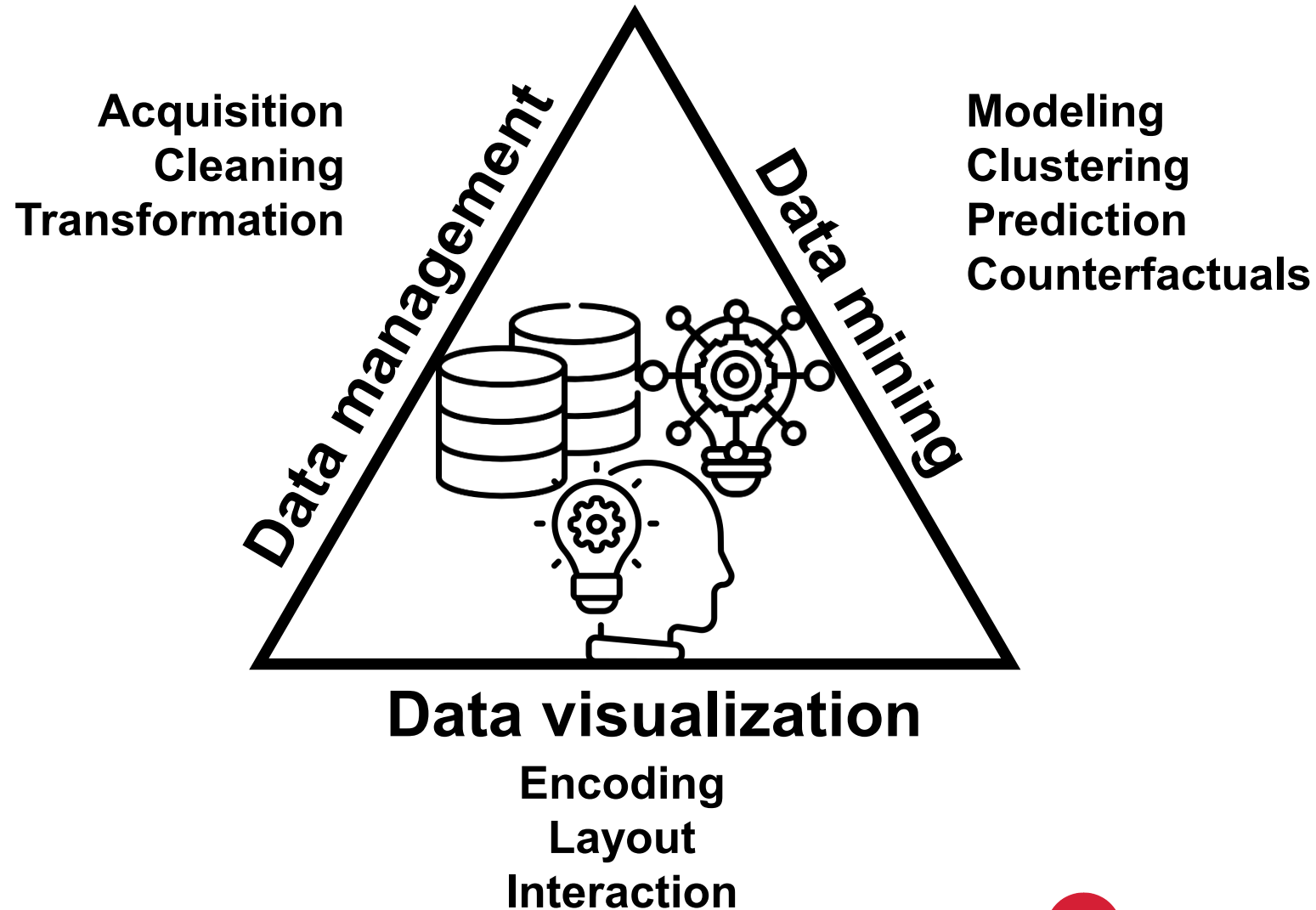
Client and server



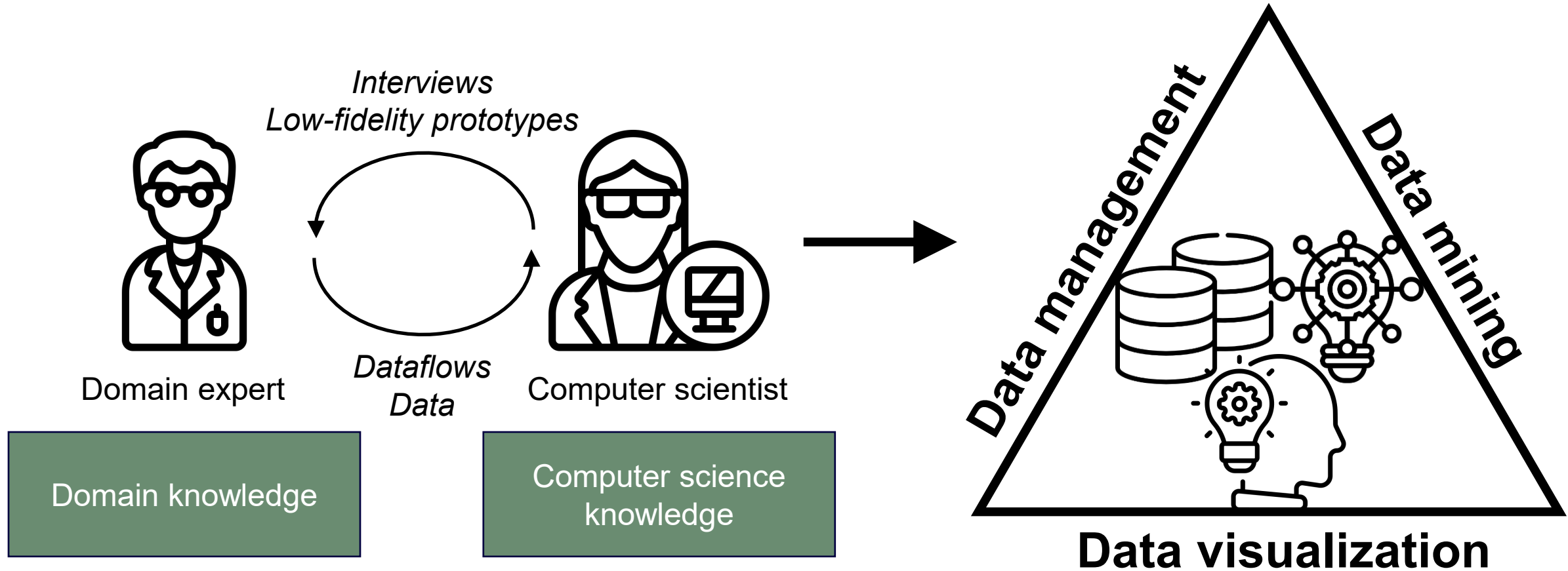
Visual Analytics



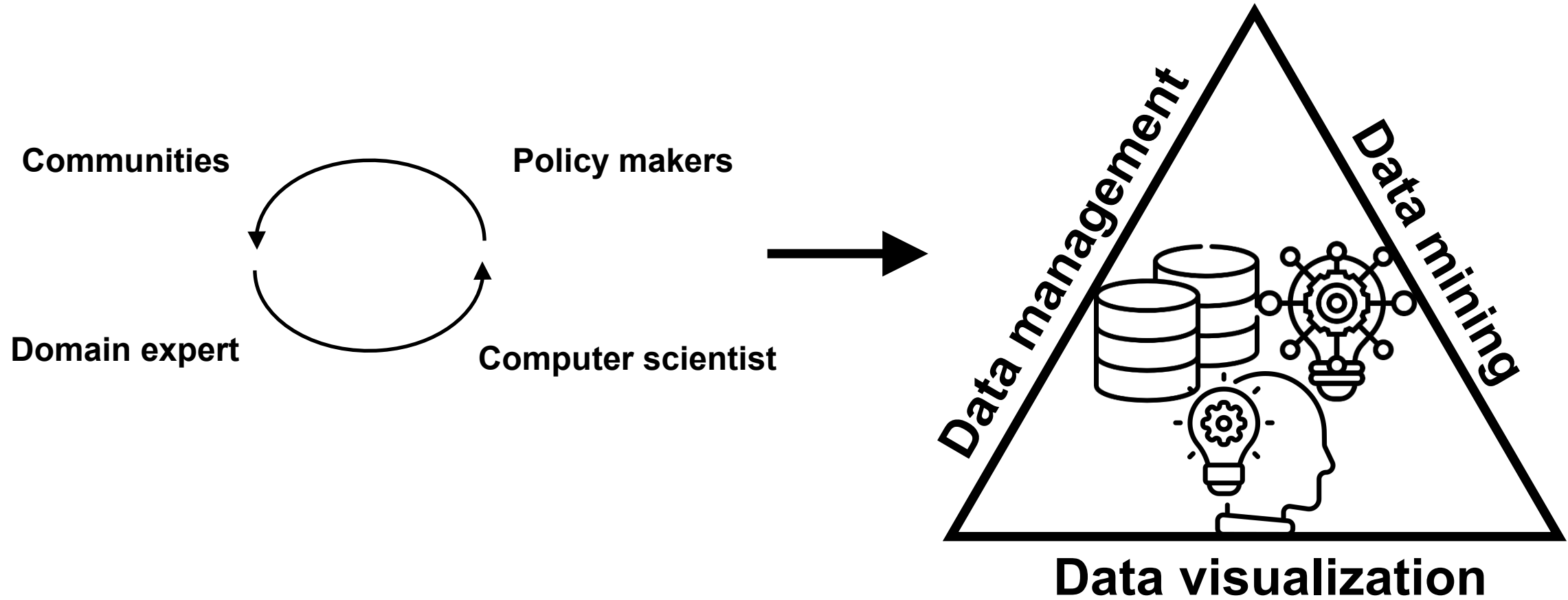
Visual Analytics



Visual Analytics



Visual Analytics



VA systems



- Attributes that distinguish well-designed VA systems:
 - **Interactivity**: latency, progressive computation
 - **Transparency**: model interpretability, uncertainty visualization
 - **Scalability**: computational and cognitive
 - **Reproducibility**: provenance tracking and re-execution

VA system dissection activity



- Small groups (3-4 members)
- Identify:
 - VA system
 - Data sources
 - Analytics methods
 - Visual representations
 - Interaction strategies
 - Potential design trade-offs

Visual Exploration of Big Spatio-Temporal Urban Data: A Study of New York City Taxi Trips

Nivan Ferreira, Jorge Poco, Huy T. Vo, Juliana Freire, and Cláudio T. Silva

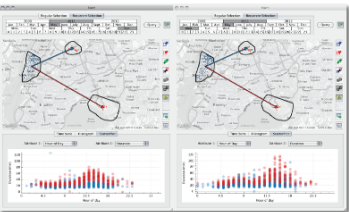


Fig. 1. Comparison of taxi trips from Lower Manhattan to JFK and LGA airports in May 2011. The query on the left selects trips that occurred on Sundays, while the one on the right selects trips that occurred on Mondays. Users specify these queries by visually selecting regions on the map and connecting them. In addition to inspecting the results depicted on the map, i.e., the dots corresponding to pickups (blue) and dropoffs (orange) of the selected trips, they can also explore the results through other visual representations. The scatter plots below the maps show the relationship between hour of the day and trip duration. Points in the plots are colored according to the spatial constraint represented by the arrows between the regions: trips to JFK in blue, and trips to LGA in red. The plots show that many of the trips on Monday between 3PM and 5PM take much longer than trips on Sundays.

Abstract—As increasing volumes of urban data are captured and become available, new opportunities arise for data-driven analysis that can lead to improvements in the lives of citizens through evidence-based decision making and policies. In this paper, we focus on a particularly important urban data set: taxi trips. Taxis are valuable sensors and information associated with taxi trips can provide unprecedented insight into many different aspects of city life, from economic activity and human behavior to mobility patterns. But analyzing these data presents many challenges. The data are complex, containing geographical and temporal components in addition to multiple variables associated with each trip. Consequently, it is hard to specify exploratory queries and to perform comparative analyses (e.g., compare different regions over time). This problem is compounded due to the size of the data—there are on average 500,000 taxi trips each day in NYC. We propose a new model that allows users to visually query taxi trips. Besides standard analytics queries, the model supports origin-destination queries that enable the study of mobility across the city. We show that this model is able to answer a wide range of analytical queries and that it is also flexible in that not only can queries be composed but also different response times; makes use of an adaptive level-of-detail rendering that shows hidden details to the users in a summary through the use of led by traffic engineers and economists that show how our model and

ment data; visual exploration

efficiently, and sustainably is among the most important undertakings in this century. While in the recent past, decision makers and social scientists faced significant constraints in obtaining the data needed to understand city dynamics and evaluate policies and practices, data are now abundant. Many cities have started to make a wide range of data sets available, see e.g., [24, 10, 7]. The challenge now is how to make sense of these data.

We examine one particularly important urban data set: taxi trips. In New York City, each day 13,000 taxis carry over one million passengers and make, on average, 500,000 trips—totaling over 170 million trips a year. Taxi trips are thus valuable sensors of city life. Consider the plot in Fig. 2, which shows how the number of trips per day varies

Urban Mosaic: Visual Exploration of Streetscapes Using Large-Scale Image Data

Fabio Miranda¹, Maryam Hosseini², Marcos Lage³, Harish Doraiswamy¹,
Graham Dove¹, Cláudio T. Silva¹

¹New York University; ²Rutgers University; ³Universidade Federal Fluminense
{fmiranda,harishd,grahamdove,cssilva}@nyu.edu; {maryhosseini@rutgers.edu; {mlage@ic.fuff.br

ABSTRACT
Urban planning is increasingly data driven, yet the challenge of designing with data at a city scale and remaining sensitive to the impact at a human scale is as important today as it was for Jane Jacobs. We address this challenge with Urban Mosaic, a tool for exploring the urban fabric through a spatially and temporally dense data set of 7.7 million street-level images from New York City, captured over the period of a year. Working in collaboration with professional practitioners, we use Urban Mosaic to investigate questions of accessibility and mobility, and preservation and retrofitting. In doing so, we demonstrate how tools such as this might provide a bridge between the city and the street, by supporting activities such as visual comparison of geographically distant neighborhoods, and temporal analysis of unfolding urban development.

Author Keywords
Urban planning; Interactive visualization; Data analysis; Urban data

CCS Concepts
•Human-centered computing → Human computer interaction (HCI); Visual analytics; Visualization toolkits;
"A sense of place is built up, in the end, from many little things too, some so small people take them for granted, and yet the lack of them takes the flavor out of the city."

include, suburbanization, economic deconcentration, modification to transport infrastructure, rezoning and/or gentrification of neighborhoods, and major renewal projects. Examples of changes at a human scale, on the other hand, are reflected in the city's urban fabric, and include aspects that make a city livable, encourage walking, and contribute to the perception of safety; in other words affect the day to day lives of its inhabitants. This might be manifested in lighting, shadow, sky exposure, open-front shops, the details of building facades, etc. [32].

Responding to these challenges at the city scale, planners and designers have been aided by a rapid growth in data from urban environments, and so are able to turn to computational methods and large-scale data analysis, which increase understanding by quantifying different aspects of the city (e.g., [36, 13, 14, 29, 70, 67, 48]). However, while sensitivity to the impact of change at a human scale remains as important today as it was for Jane Jacobs and others in the 1950s and 1960s [47], analyses of suitable data, which emphasize qualitative, visual details, are often difficult and time-consuming to perform. Although there is an increasing availability of street-view images, which support a degree of virtual assessment and auditing of the built environment, their distribution is often temporally sparse and so analysis is limited.

This paper introduces Urban Mosaic, a tool for visually exploring the urban fabric. It responds to the challenges practitioners face by employing a newly available spatially and temporally dense data set of street-level images from New York City (NYC). Urban Mosaic is a visual exploration system designed to help practitioners in urban planning and design gain insight into the human scale impact of changes in the urban fabric. It utilizes state-of-the-art computer vision techniques for image similarity search and clustering, together with efficient spatio-temporal selection and aggregation over the image metadata to visually explore and map this image data set. It further allows the analysis to be augmented using spatio-temporal urban data from a variety of other sources (e.g., census, transport, crime, weather, housing market, zoning, noise complaints). The image data set used in this work contains 7.7 million images captured in the Manhattan and Brooklyn boroughs between April 2016 and April 2017 using car-mounted cameras. Urban Mosaic has been developed as a collaboration between researchers in urban planning, visual analytics, and HCI. We include a detailed

HORA 3D: Personalized Flood Risk Visualization as an Interactive Web Service

Silvana Rauer-Zachmeister¹, Daniel Cornet¹, Bernhard Stadler¹, Zoltan Horvath¹, Artem Konev¹,
Andreas Buttinger-Kreuthuber¹, Raimund Heidrich², Günter Hölzl³, Eduard Gröller⁴, and Jürgen Wauer¹

¹VRVis Zentrum für Virtual Reality und Visualisierung, Forschungsbereich, Vienna, Austria
²ROCOM Engineering Office for Environmental Engineering and Water Management, Vienna, Austria
³Institute of Hydraulic Engineering and Water Resource Management, TU Wien, Vienna, Austria
⁴Institute of Visual Computing and Human-Centered Technology, TU Wien, Vienna, Austria

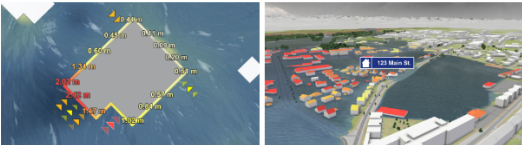


Figure 1: Flood risk visualization of a selected building of interest, highlighting water depths around the building and high impact velocities on exterior walls via animated arrows in top-down view (left). High water depths at buildings in the vicinity are emphasized via roof coloring in overview (right).

Abstract
We propose an interactive web-based application to inform the general public about personal flood risks. Flooding is the natural hazard affecting most people worldwide. Protection against flooding is not limited to mitigation measures, but also includes communicating its risks to affected individuals to raise awareness and preparedness for its adverse effects. Until now, this is mostly done with static and indiscriminate 2D maps of the water depths. These flood hazard maps can be difficult to interpret on prior knowledge. In addition to the hazard, the flood risk has to high water depths and flow velocities as well as the vulnerability of structures personalized visualization to raise awareness of their risk. It shows only the relevant information from large precomputed flood simple and comprehensible. To achieve this goal, we extend various visualization and interaction techniques in 3D. A new view-dependent interactive interpretation of the visualization to perform predefined inform themselves about their flood risks. We evaluated the user points of different levels of expertise, who rated the helpfulness of the

nation systems → Spatial-temporal systems;

CGF 43-3 (2024)

PROWIS: A Visual Approach for Building, Managing, and Analyzing Weather Simulation Ensembles at Runtime

Carolina Veiga Ferreira de Souza, Suzanna Maria Bonnet,
Daniel de Oliveira, Marcio Cataldi, Fabio Miranda, and Marcos Lage

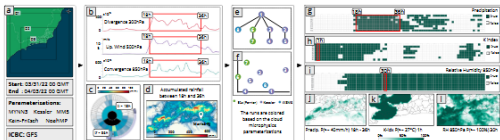


Fig. 1. Building, managing, and analyzing a Weather Research and Forecasting (WRF) model ensemble using PROWIS to study an extreme rainfall event in Maricá (RJ, Brazil) in April 2022. Before running the first simulation, weather experts defined the simulation domain, a 72-hour time horizon, the initial and boundary conditions data, and a set of physical parameterizations (a). After running the first simulation, using the line chart panel, they identified relevant predictions of divergence at 500 hPa, convergence at 850 hPa, and vertical upward at 500 hPa between 18h and 30h (b). By selecting this time interval in the scatter plot (c), they saw that the model predicted heavy rain for several locations but underestimated the volume in the city (white marker in (d)). Using the runs overview graph (e), the experts generated a simulation ensemble for a broader study. No pattern among the members was observed in the rain volume results (f). The experts then inspected the heat maps to verify the chances of having a minimum rainfall of 40 mm between 18h and 30h (g). However, in this period, the probability of this scenario was small in Maricá (h). Similarly, a minimum K-index of 27 (i) until 27h (j) and relative humidity of 100% (k), two conditions related to extreme rainfall, could be observed only outside the city (l). The results show that the model underestimated the events' disaster potential in Maricá.

Abstract—Weather forecasting is essential for decision-making and is usually performed using numerical modeling. Numerical weather models, in turn, are complex tools that require specialized training and laborious setup and are challenging even for weather experts. Moreover, weather simulations are data-intensive computations and may take hours to days to complete. When the simulation is finished, the experts face challenges analyzing its outputs, a large mass of spatiotemporal and multivariate data. From the simulation setup to the analysis of results, working with weather simulations involves several manual and error-prone steps. The complexity of the problem increases exponentially when the experts must deal with ensembles of simulations, a frequent task in their daily duties. To tackle these challenges, we propose PROWIS: an interactive and provenance-oriented system to help weather experts build, manage, and analyze simulation ensembles at runtime. Our system follows a human-in-the-loop approach to enable the exploration of multiple atmospheric variables and weather scenarios. PROWIS was built in close collaboration with weather experts, and we demonstrate its effectiveness by presenting two case studies of rainfall events in Brazil.

Index Terms—Weather visualization, Ensemble visualization, Provenance management, WRF visual setup

perform weather predictions, climate specialists must consider an enormous amount of data (e.g., from satellites, weather stations), construct ensembles of numerical simulations, and rely on past experience. Ensembles of numerical weather simulations are computed using mathematical models that describe atmospheric behavior through equations based on physical laws [31]. These models depend on initial conditions, terrain characterization, parameterization of physical processes, and discretization strategies. Weather experts must go through dependent and complex steps to run a simulation. These steps and their data dependencies resemble large-scale scientific workflows [8]. First, meteorologists must define the simulation domain considering terrain non-orthogonality, land-use information, and spatial discretization aspects. Then, they configure the time horizon and discretization to be used during the simulation. Based on the spatiotemporal setup, the user must set the initial and boundary atmospheric conditions. Finally, previous steps are considered to specify the dynamical/thermodynamic behaviors and the micro-scale processes.

Usually, setting up these steps requires editing large text files to define several parameters. Also, to run each step, the weather expert must

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VA system dissection activity

- Small groups (3-4 members)
- Identify:
 - VA system
 - Data sources
 - Analytics methods
 - Visual representations
 - Interaction strategies
 - Potential design trade-offs

